

updated: 4/26

# Curriculum Vitae: Chad McKell

## ABOUT

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| Position  | Research Engineer / Ph.D. Candidate                                                                                                                                                                                                                                                                                                                                                                                           |
| Company   | Sigma Connectivity, on assignment at Meta                                                                                                                                                                                                                                                                                                                                                                                     |
| Address   | 10301 Willows Road NE, Redmond, WA 98052                                                                                                                                                                                                                                                                                                                                                                                      |
| Email     | cmckell@meta.com                                                                                                                                                                                                                                                                                                                                                                                                              |
| Website   | chadmckell.com                                                                                                                                                                                                                                                                                                                                                                                                                |
| Languages | English (native), Spanish (B2), French (A2)                                                                                                                                                                                                                                                                                                                                                                                   |
| Code      | Python, C++/CUDA, MATLAB, LaTeX                                                                                                                                                                                                                                                                                                                                                                                               |
| Research  | My research spans mathematical modeling and numerical simulation of physical systems, with an emphasis on acoustics and general wave physics. Recent projects include conformal symmetry for exterior wave problems; differential forms for non-spherical harmonic decomposition; and variational principles for elastic wave modeling. Past projects include underwater acoustic sensing and optical standing-wave trapping. |

## EDUCATION

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| 9/19–(exp. 9/26) | <b>University of California San Diego</b> , Ph.D. in Computer Music<br>Research area: Computational Acoustics and Applied Mathematics<br>Dissertation: <i>Conformal Symmetry for Exterior Wave Problems</i><br>Co-advisors: Albert Chern (CSE), Miller Puckette (Music; IRCAM) |
| 9/16–10/17       | <b>University of Edinburgh</b> , M.S. in Acoustics and Music Technology                                                                                                                                                                                                        |
| 8/09–12/15       | <b>Wake Forest University</b> , M.S. in Physics                                                                                                                                                                                                                                |
| 6/02–8/09        | <b>Brigham Young University</b> , B.S. in Biophysics                                                                                                                                                                                                                           |

## EMPLOYMENT

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| 9/24–      | <b>Sigma Connectivity, on assignment at Meta</b> , Research Engineer                   |
| 3/24–6/24  | <b>University of California San Diego</b> , Teaching Assistant in Computer Science     |
| 6/23–2/24  | <b>Meta, Reality Labs Research</b> , Research Scientist Intern                         |
| 8/21–3/22  | <b>Meta, Reality Labs Research</b> , Research Intern                                   |
| 9/19–6/23  | <b>University of California San Diego</b> , Teaching Assistant/Researcher in Music     |
| 7/18–7/19  | <b>Applied Research in Acoustics</b> , R&D Scientist                                   |
| 5/18–5/18  | <b>Moog Music (self-employed)</b> , Audio Software Developer                           |
| 4/17–9/17  | <b>Lofelt (self-employed)</b> , Acoustics Researcher                                   |
| 10/14–8/16 | <b>J.P. Morgan/Neovest (via ConsultNet)</b> , Software Development Engineer in Test    |
| 8/12–12/12 | <b>University of North Carolina School of the Arts</b> , Adjunct Instructor of Physics |
| 9/09–9/11  | <b>Wake Forest University</b> , Teaching Assistant in Physics                          |
| 9/08–6/09  | <b>Brigham Young University</b> , Tutorial Lab Assistant in Physics                    |
| 8/07–3/09  | <b>Brigham Young University</b> , Research Assistant in Philosophy                     |

## RESEARCH ACTIVITIES

|       |                                                                                                                                                                                                                                                                                                                                                                                  |
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| 9/24– | <b>Sigma Connectivity, on assignment at Meta</b> , Research Engineer<br>Redmond, Washington. Affiliation: Reality Labs Research Audio. Research area: <i>spatial audio</i> . Build novel spatial audio technology for Reality Labs Research at Meta. I work onsite at Meta as a vendor employed by Sigma Connectivity. Supervisor: Sebastian Prepeliță. Team Lead: Ravish Mehra. |
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## RESEARCH ACTIVITIES CONT.

- 9/19–9/24      **University of California San Diego**, Ph.D. Student  
La Jolla, California. Affiliation: Center for Visual Computing; Department of Music. Research area: *computational acoustics, applied mathematics*. Developed geometric and numerical methods for solving exterior wave problems, with applications in virtual acoustics and related fields. Committee: Albert Chern (co-chair, CSE), Miller Puckette (co-chair, Music; IRCAM), Melvin Leok (Mathematics), Stefan Bilbao (IRCAM), Sebastian Prepelîță (Meta), Shahrokh Yadegari (Music).
- 6/23–2/24      **Meta, Reality Labs Research**, Research Scientist Intern  
Redmond, Washington. Affiliation: Reality Labs Research Audio. Research area: *spatial audio*. Built novel spatial audio technology for Reality Labs Research at Meta. Supervisor: Sebastian Prepelîță. Team Lead: Ravish Mehra.
- 8/21–3/22      **Meta, Reality Labs Research**, Research Intern  
La Jolla, California. Research description: see above.
- 7/18–7/19      **Applied Research in Acoustics**, R&D Scientist  
Culpeper, Virginia. Research area: *underwater acoustic sensing, acoustic beamforming*. Developed physics-based signal processing algorithms for object detection and classification in naval sonar systems. Team Lead: Jonathan Botts.
- 1/17–8/17      **University of Edinburgh**, Master’s Student  
Edinburgh, Scotland. Affiliation: Acoustics and Audio Group, Reid School of Music. Research area: *computational acoustics, numerical simulation*. Developed numerical simulations for commercial audio-haptic devices, including the Razer Nari Ultimate headphones. Received research funding from Lofelt, a Berlin-based haptics startup acquired by Meta in 2022. Advisor: Stefan Bilbao.
- 1/10–9/13      **Wake Forest University**, Master’s Student  
Winston-Salem, North Carolina. Affiliation: Department of Physics. Research area: *optical trapping, laser beam characterization, fluid dynamics*. Designed, constructed, and analyzed standing-wave optical traps for Brownian particle tracking. Published work in the Journal of the Optical Society of America B. Advisor: Keith Bonin.
- 8/07–8/09      **Brigham Young University**, Undergraduate Student  
Provo, Utah. Affiliation: Department of Physiology and Developmental Biology. Research area: *biophysics, structural biology*. Investigated structural effects of isoflurane on artificial lipid bilayers using atomic force microscopy. Advisor: David Busath.

## TEACHING EXPERIENCE

### As Instructor

**UNCSA**  
SCI 1100      General Physics. Fall 2012 (1 term).

## TEACHING EXPERIENCE CONT.

### As TA

#### UCSD

|         |                                                        |
|---------|--------------------------------------------------------|
| CSE 291 | Physics Simulation. Spring 2024 (1 term).              |
| MUS 5   | Sound in Time. Spring 2020 (1 term).                   |
| MUS 6   | Electronic Music. Fall 2020 (1 term).                  |
| MUS 15  | Popular Music: David Bowie. Winter 2021 (1 term).      |
| MUS 15  | Popular Music: Video Game Music. Winter 2020 (1 term). |
| MUS 171 | Computer Music I. Winter 2022 (1 term).                |
| MUS 172 | Computer Music II. 2021–2022 (2 terms).                |

#### WFU

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|---------|-----------------------------------------------------|
| PHY 113 | General Physics I (Mechanics). 2009–2011 (4 terms). |
| PHY 114 | General Physics II (E&M). Fall 2010 (1 term).       |

### As Tutor

#### BYU

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|-----------|--------------------------------------------------------------|
| PHSCS 105 | General Physics 1 (Mechanics). 2008–2009 (2 terms).          |
| PHSCS 106 | General Physics 2 (E&M). Winter 2009 (1 term).               |
| PHSCS 121 | Principles of Physics 1 (Mechanics). 2008–2009 (2 terms).    |
| PHSCS 123 | Principles of Physics 2 (Waves/Thermo). W/Sp 2009 (2 terms). |
| PHSCS 220 | Principles of Physics 3 (E&M). W/Sp 2009 (2 terms).          |

## PUBLICATIONS

### Manuscripts in Progress

- (1) **C. McKell**, M. Nabizadeh, S. Wang, and A. Chern, “Wave simulations in infinite spacetime”.

*Simulating wave propagation on an infinite domain has been a long-standing computational challenge. Conventional approaches to this problem only produce wave simulations on a small subset of the infinite domain. Using the fact that wave propagation on an infinite Minkowski spacetime is equivalent to wave propagation on a bounded Minkowski spacetime under a Kelvin-like transformation, we simulate wave propagation on the entire infinite domain using a finite discretization of the bounded domain with no additional loss of accuracy from the transformation.*

- (2) **C. McKell**, *Conformal Symmetry for Exterior Wave Problems*, Ph.D. Dissertation, University of California San Diego. Defense planned for Fall 2025. Advisors: Albert Chern and Miller Puckette.

*This work presents novel geometric methods for handling open exterior boundaries in wave simulations. First, I discuss extensions to the reflectionless discrete perfectly matched layer for the wave and Helmholtz equations. Then, I demonstrate that the conformal invariance of the wave equation under a Kelvin transform in Minkowski spacetime allows one to convert an infinite domain problem into a bounded domain problem that can be solved using standard numerical methods with no additional loss of accuracy introduced by the transform. I conclude by discussing parallel computing strategies for the geometric boundary models, applications in architectural acoustics and binaural audio, and future work in obstacle boundary flattening.*

## PUBLICATIONS CONT.

### Journal Articles

- (3) **C. McKell** and K. Bonin, “Optical corral using a standing-wave Bessel beam,” *Journal of the Optical Society of America B*, Vol. 35, No. 8, 1910–1920, 2018.

### Conference Proceedings

- (4) **C. McKell**, “Sonification of optically-ordered Brownian motion,” In Proceedings of the International Computer Music Conference (ICMC), Utrecht, Netherlands, September 2016.

### Master’s Theses

- (5) **C. McKell**, *Real-Time Physical Modeling for Haptic Feedback Rendering*, Master’s Thesis, University of Edinburgh, Acoustics and Audio Group, 2017. Advisor: Stefan Bilbao.
- (6) **C. McKell**, *Confinement and Tracking of Brownian Particles in a Bessel Beam Standing Wave*, Master’s Thesis, Wake Forest University, Department of Physics, 2015. Advisor: Keith Bonin.